Hojin Chu

CURRICULUM VITAE
Updated October 28, 2026

Research Fellow, Korea Institute for Advanced Study (KIAS) Mar. 2025 – Present
School of Computational Sciences • Mentor: Jeong Han Kim

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Research Interests

Graph Theory, Combinatorics, Discrete Mathematics, and Combinatorial Matrix Theory
Particular interests: Extremal and Structural Graph Theory

Education

Seoul National University Seoul, Korea
Ph.D. in Mathematics — Mar. 2020 – Feb. 2025
Thesis: *A study of digraph structure in terms of common neighbors*
Advisor: **Suh-Ryung Kim**

B.S. in Mathematics Education — Mar. 2016 – Feb. 2020

Fellowships & Grants

Basic Science Research Program — June 2022 – May 2023
National Research Foundation of Korea • KRW 20,000,000/year
Role: **Principal Investigator (PI)**
Project title: Competition graphs of degree bounded digraphs

Military Service

Completed; duty period: 3 years
Alternative service at KIAS — Mar. 2025 – Aug. 2026
Alternative service at Seoul National University — Sept. 2023 – Feb. 2025

Published Papers

4. **Digraphs in which every t vertices have exactly λ common out-neighbors**
Myungho Choi, **Hojin Chu**, and Suh-Ryung Kim.
Discrete Mathematics 348(10) (2025), 114580. DOI: 10.1016/j.disc.2025.114580

3. **Structural properties of symmetric Toeplitz and Hankel matrices**
Hojin Chu and Homoon Ryu.
Linear Algebra and its Applications 708 (2025), 204–216. DOI: 10.1016/j.laa.2024.11.025

2. **A digraph version of the Friendship Theorem**
Myungho Choi, **Hojin Chu**, and Suh-Ryung Kim.
Discrete Mathematics 348(1) (2025), 114238. DOI: 10.1016/j.disc.2024.114238

1. **Competition graphs of degree bounded digraphs**
Hojin Chu and Suh-Ryung Kim.
Discrete Applied Mathematics 343 (2024), 106–114. DOI: 10.1016/j.dam.2023.10.009

Submitted Papers & Preprints

12. **On strong odd colorings of sparse graphs**
Ilkyoo Choi and **Hojin Chu**. Preprint, 2026. arXiv:2610.34874

11. **Connectivity keeping paths in digraphs**
Hojin Chu, Boram Park, and Homoon Ryu. Preprint, 2026. arXiv:2608.25240

10. **A sharp extension of Halin's removable-edge theorem to matchings**
Hojin Chu. Preprint, 2026. arXiv:2608.09394

9. **Minimum degree conditions for removable matchings in k-connected graphs**
Hojin Chu, Ringi Kim, and Boram Park. Preprint, 2026. arXiv:2607.17533

8. **On 2-connected graphs without cycles of length 1 modulo 3**
Yandong Bai, **Hojin Chu**, Binlong Li, Boram Park, and Homoon Ryu. Preprint, 2026. arXiv:2606.02356

7. **Existence of cycles of length divisible by 3 or 4**
Ilkyoo Choi, **Hojin Chu**, Ringi Kim, and Boram Park. Preprint, 2026. arXiv:2605.02731

6. **Connectivity keeping trees in triangle-free graphs**
Hojin Chu, Shinya Fujita, Boram Park, and Homoon Ryu. Preprint, 2025. arXiv:2511.06622

5. **On 2-connected graphs avoiding cycles of length 0 modulo 4**
Hojin Chu, Boram Park, and Homoon Ryu. Preprint, 2025. arXiv:2507.12798

4. **Linear-time computation of the Frobenius normal form for symmetric Toeplitz matrices via graph-theoretic decomposition**
Hojin Chu and Homoon Ryu. Preprint, 2025. arXiv:2505.20811

3. **Competition-common enemy graphs of degree-bounded digraphs**
Myungho Choi, **Hojin Chu**, and Suh-Ryung Kim. Preprint, 2024. arXiv:2405.13363

2. **Digraphs in which every t vertices share exactly λ out-neighbors and exactly λ in-neighbors**
Hojin Chu and Suh-Ryung Kim. Preprint, 2024. arXiv:2405.13293

1. **Planarity of generalized ladder graphs**
Hojin Chu, Suh-Ryung Kim, and Homoon Ryu. Preprint, 2022. arXiv:2208.13637

Research Talks

International

[Contributed] SIAM Conference on Discrete Mathematics (DM26)
University of California San Diego, San Diego, United States June 22–25, 2026
Title: Connectivity keeping trees in triangle-free graphs

[Contributed] DMA 2026 (International Workshop on Discrete Mathematics and Algorithms 2026)
Kunibiki Messe, Matsue, Japan Mar. 16–18, 2026
Title: On 2-connected graphs avoiding cycles of length 0 modulo 4

[Minisymposium] ILAS 2025 (The 26th Conference of the International Linear Algebra Society)
National Sun Yat-sen University, Kaohsiung, Taiwan June 23–27, 2025
Title: Structural properties of symmetric Toeplitz and Hankel matrices

[Contributed] CanaDAM 2025 (Canadian Discrete and Algorithmic Mathematics Conference)
University of Ottawa, Ottawa, Canada May 20–23, 2025
Title: Directed analogues of Friendship graphs

Domestic

[Contributed] 2026 Combinatorics Workshop
Konjiam Resort, Gwangju-si, Korea Oct. 21, 2026
Title: Minimum degree conditions for removable matchings in k-connected graphs

[Invited] Combinatorics Seminar, Hanyang University
Hanyang University, Seoul, Korea Oct. 6, 2026
Title: A sharp extension of Halin's removable-edge theorem to matchings

[Invited] One-Day Workshop on Graph Theory
Seoul National University, Seoul, Korea Aug. 21, 2026
Title: Cycles modulo 3 and 4: Extremal and Structural results

[Invited] 35th KIAS Combinatorics Workshop
Shilla Stay Haeundae, Busan, Korea Dec. 22–24, 2025
Title: Connectivity keeping subgraph deletion problem

[Invited] KPPY 100 (Young Seminar)
Kolon Hotel, Gyeongju, Korea Sept. 19–21, 2025
Title: On 2-connected graphs avoiding cycles of length 0 modulo 4

[Invited] 2025 Korean Student Combinatorics Workshop
The-K Hotel, Gyeongju, Korea Aug. 20–24, 2025
Title: Things I worry about as a Postdoc

[Invited] SUNY Korea Combinatorics Seminar
SUNY Korea, Incheon, Korea July 15, 2025
Title: Directed analogues of Friendship graphs

[Contributed] 2024 Fall KMS (Korean Mathematical Society) Conference
Sungkyunkwan University, Suwon, Korea Oct. 24–26, 2024
Title: Digraphs in which every t vertices share exactly λ out-neighbors and exactly λ in-neighbors

[Contributed] 2023 Fall KMS (Korean Mathematical Society) Conference
Seoul National University, Seoul, Korea Oct. 26–28, 2023
Title: Competition graphs of degree bounded digraphs

[Invited] AORC (Applied Algebra and Optimization Research Center) Seminar
Sungkyunkwan University, Suwon, Korea May 26, 2023
Title: A digraph version of the Friendship Theorem

Professional Service

Conference Organization

Co-organizer **DMA 2027**
International Workshop on Discrete Mathematics and Algorithms 2027
Jeju Shinhwa World, Jeju, Korea Mar. 16–20, 2027

Refereeing

SIAM Journal on Discrete Mathematics; Discrete Mathematics; Discrete Applied Mathematics;
Discussiones Mathematicae Graph Theory; Ars Combinatoria; FOCS 2026.

Public Talks

Public Lecture

AI & Math Academy
Organized by KIAS & Dongdaemun-gu Office
KIAS, Seoul, Korea — Feb. 5, 2026
Title: Discovering hidden connections in the World through Graphs
(그래프로 배우는 세상 속 숨은 연결 찾기)

KIAS Outreach Program

Visit from Baewha Girls' High School
KIAS, Seoul, Korea — Sept. 3, 2026
Title: Choosing a Path in Mathematics: Research, Careers, and Opportunities

Visit from UNIST Mathematics & Physics Club "Elite of Elites"
KIAS, Seoul, Korea — Sept. 26, 2025
Title: Careers in Mathematics: What it means to do Mathematical Research

Visit from Mokpo Deogin High School
KIAS, Seoul, Korea — Sept. 5, 2025
Title: A Mathematician's Career Path and Research

Teaching Experience

Seoul National University Teaching Assistant

Graph Theory	Fall 2024
Linear Algebra II	Fall 2024
Discrete Mathematics	Spring 2024
Linear Algebra II	Fall 2023
Abstract Algebra I	Spring 2023
Number Theory	Fall 2022
Abstract Algebra II	Fall 2022
Abstract Algebra I	Spring 2022
Abstract Algebra II	Fall 2021